

04 — Demonstration Protocol & Verification Record



DOCUMENT PURPOSE

Defines the live assessment procedure: the demonstration environment, the preconditions verified before assessment begins, the ordered sequence of demonstrations with the evidence produced by each, the contingency for component failure, and an **independent verification sheet** on which the assessor records results. Every capability claim in this documentation set is demonstrated by a step in this protocol and recorded on that sheet.

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1. Assessment objectives

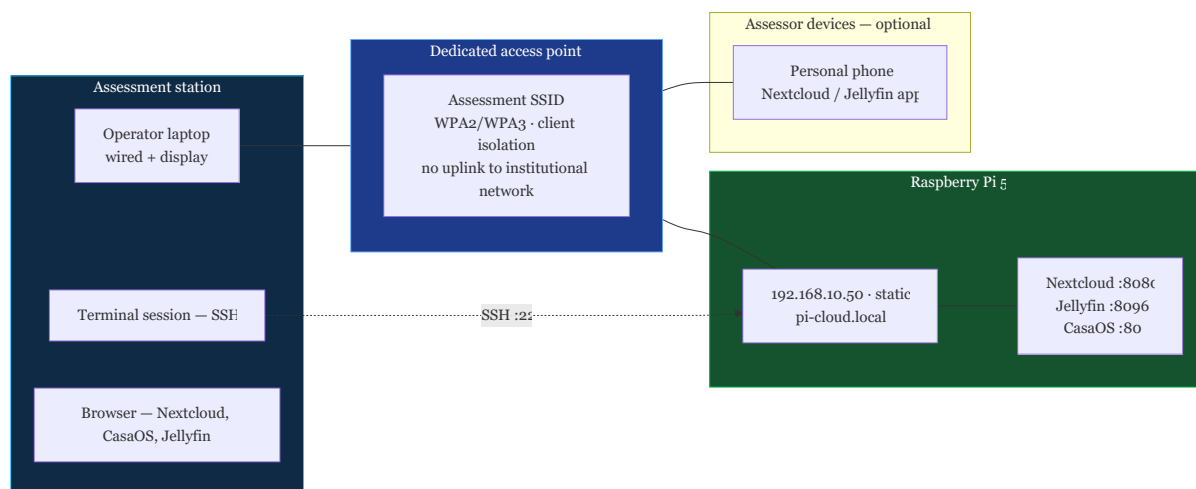
WHAT THE ASSESSMENT MUST ESTABLISH

#	Objective	Demonstrated by
O1	The system is functional — file storage, multi-user accounts, and media streaming operate correctly	D-1, D-3, D-4
O2	Data is genuinely encrypted at rest , verified by independent inspection rather than application reporting	D-2
O3	Users are isolated from one another	D-5
O4	The system is independent of internet connectivity	D-6
O5	The system recovers unattended from a full restart	D-7
O6	Data is recoverable — a backup is restored and opened	D-8
O7	The build is reproducible — the deployment is declared in version-controlled files, and recovery does not require re-authoring	D-9
O8	Costs and limits are stated accurately	Documents 01 §9, 02 §11, 05

DESIGN PRINCIPLE OF THIS PROTOCOL

Demonstrations are ordered so that **the strongest verification is performed first** and the most operationally fragile demonstration last. D-2 (filesystem inspection) and D-8 (restore) require a working terminal and take the longest; they are positioned where a subsequent failure cannot invalidate them.

2. Demonstration environment



ASSESSMENT NETWORK

The system is demonstrated on a **dedicated access point with no uplink to the institutional network**. This satisfies the isolation requirement (control 12, document 02 §10) and removes any dependency on external connectivity during the assessment. A distinct subnet is used so that a pre-configured address on a different network cannot conflict with the demonstration configuration.

Component	Configuration	Rationale
Access point	Dedicated, no uplink, 2.4 GHz + 5 GHz, client isolation enabled	Segregates assessment traffic; prevents device-to-device reachability
Host address	192.168.10.50 , static	Non-overlapping with the operational deployment range; documented on all reference material
Host name	pi-cloud.local (mDNS)	Address-independent reference; the same name resolves on the operational network
Operator station	Wired Ethernet	Removes wireless variability from the assessment
Demonstrator client	A personal phone or tablet, optional	D-11 demonstrates third-party client compatibility; it is optional because the mobile app requires prior installation by the assessor

WHY ASSESSOR DEVICES ARE OPTIONAL RATHER THAN ASSUMED

Installing a client application onto an assessor's personal device is inappropriate as a precondition. D-1 through D-10 are complete and self-contained on the operator station. D-11 is executed only if the assessor has the application already available, and is recorded as *not performed* otherwise. **A demonstration is not marked as failed because a precondition outside the project's control was not met.**

3. Network configuration

TWO DOCUMENTED AND COMMON FAILURE MODES

Symptom	Cause	Resolution
Host unreachable, but responds to <code>ping</code>	Wireless client isolation enabled on the access point	Disable client isolation, or connect the host by cable
Host unreachable entirely	Static address in a subnet the assessment laptop is not on	Confirm the laptop is associated with the assessment SSID and holds an address in <code>192.168.10.0/24</code>

Both were encountered during preparation of this deployment and both produce identical external symptoms. The readiness check in §4 detects each before assessment begins.

Address reference

Service	URL
Management dashboard	<code>http://192.168.10.50</code>
File service	<code>http://192.168.10.50:8080</code>
Media service	<code>http://192.168.10.50:8096</code>
Host name (any service)	<code>http://pi-cloud.local[:8080 \ :8096]</code>
Administrative access	<code>ssh <user>@192.168.10.50</code>

4. Pre-assessment readiness check

PROCEDURE

Executed **30 minutes before assessment begins** and repeated immediately before. Every row must read PASS. A single FAIL aborts the assessment and is remediated before it proceeds — a partially functional system presented as complete is a worse outcome than a delayed start.

picloud-health

#	Check	Method	Required result
R1	Host reachable	<code>ping -c3 192.168.10.50</code>	0% packet loss
R2	Hostname resolves	<code>ping -c2 pi-cloud.local</code>	Replies
R3	Compute platform	<code>lscpu \\\ head -5</code>	4 × Cortex-A76, aarch64
R4	Memory	<code>free -h</code>	≈7.6 GB total, low swap usage
R5	Storage present	<code>findmnt /mnt/storage</code>	Mounted on <code>/mnt/storage</code>
R6	Storage capacity	<code>df -h /mnt/storage</code>	> 20% free
R7	Storage health	<code>smartctl -H /dev/nvme0n1</code>	PASSED
R8	Power / thermal state	<code>vcgencmd get_throttled</code>	<code>throttled=0x0</code>
R9	Temperature	<code>vcgencmd measure_temp</code>	< 70 °C
R10	Docker root location	<code>docker info \\\ grep "Docker Root Dir"</code>	<code>/var/lib/docker</code> (microSD)
R11	Container state	<code>docker ps --format '{{.Names}} {{.Status}}'</code>	All services <code>Up</code> , no <code>Restarting</code>
R12	CasaOS dashboard	Browse <code>http://192.168.10.50</code>	Renders
R13	Nextcloud	Browse <code>http://192.168.10.50:8080</code>	Login page, no error banner
R14	Jellyfin	Browse <code>http://192.168.10.50:8096</code>	Dashboard loads
R15	Encryption state	<code>php occ encryption:status</code>	<code>enabled</code> , per-user
R16	Background jobs	<code>crontab -u www-data -l</code>	5-minute entry present

#	Check	Method	Required result
R17	Self-registration	<code>php occ config:app:get core registration.enabled</code>	<code>false</code>
R18	Snapshots current	<code>ls -lt /mnt/storage/snapshots \ head -1</code>	Timestamp within the last 2 hours
R19	Off-site copy present	<code>ls -l /mnt/backup-usb/nextcloud-data</code>	Non-empty, dated within 7 days
R20	Log rotation	<code>docker inspect jellyfin --format '{{.HostConfig.LogConfig}}'</code>	<code>json-file</code> , size limited
R21	MicroSD headroom	<code>df -h /</code>	< 70% used
R22	Demonstration accounts	Manual check	Test accounts present, credentials known
R23	Test artefacts	Manual check	Demonstration file present with a known plaintext string
R24	Restore test file	Manual check	A known file exists on the off-site copy for D-8

R23 IS A SPECIFIC AND NECESSARY PREPARATION STEP

The encryption demonstration (D-2) works by writing a file containing a known string, then searching the storage volume for that string. **The string must be prepared in advance** — for example `PI-CLOUD-VERIFY-20260930` . Without it, the demonstration degrades into an assertion, which is the outcome this protocol exists to avoid.

5. Demonstration sequence

ORDERING RATIONALE

Total duration $\approx$ 22 minutes. D-1 $\rightarrow$ D-2 establishes function and encryption immediately. D-6 (internet disconnection) and D-8 (restore) are the two procedures with real elapsed time and are positioned at D-6 and D-8, after which no subsequent step depends on external connectivity.

D-1 — Functional overview (≈ 2 min)

Claim: the system is operational and provides three distinct service classes.

Procedure

1. CasaOS dashboard — display container state, CPU, memory, and both storage volumes. *Evidence: control 1 — the management plane reports actual resource state rather than a static display.*
2. Nextcloud web interface — sign in, open the Files view.
3. Jellyfin — open the library and display the media index.
4. Samba — open the share from a file manager on the operator laptop (`\\192.168.10.50\family`).

Result: three service classes reachable from standard clients, with no client software required for the fourth.

D-2 — Encryption at rest, verified by inspection (≈ 3 min)

✓ PRIMARY VERIFICATION EVENT

This is the demonstration that substantiates the central technical claim, and it is performed by **direct filesystem inspection rather than by reading the application's report of its own configuration.**

Procedure

```
# 1. Confirm the configured state
cd /mnt/storage/nextcloud-data
sudo -u www-data php occ encryption:status
```

```
# 2. Write a file containing a known plaintext string through the web interface
# (upload: PI-CLOUD-VERIFY-20260930.txt)
```

```
# 3. Search the storage volume for that string
sudo grep -rl "PI-CLOUD-VERIFY-20260930" /mnt/storage/nextcloud-data/data
```

```
# 4. Inspect the raw stored content
sudo -u www-data php occ files:scan --all
find /mnt/storage/nextcloud-data -name 'PI-CLOUD-VERIFY-20260930.txt' \
```

```
-exec sh -c 'echo "== {} =="; head -c 64 "{}" | xxd | head -4' \;
```

Required results

Step	Required result
3	No output. Any filename returned indicates unencrypted storage
4	Binary output with no readable text

Conclusion supported: the content visible through the application is not present in readable form on the storage medium.

Limitation to state at this step: this protects data at rest. It does not prevent an administrator with a live authenticated session from reading a file while its owner is signed in. That boundary is specified in document 02 §7.4.1 and is not concealed by this demonstration.

D-3 — Multi-user operation and isolation (*≈2 min*)

Claim: distinct accounts operate independently and are isolated.

Procedure

1. Sign in to Nextcloud as user **A**; create a folder and upload a file.
2. Sign in as user **B**.
3. Attempt to navigate to user A's folder.
4. Compare per-user quota reporting between the two accounts.

Required result: the cross-user access attempt is **denied**, and the interruption is presented as a permissions error rather than an application failure. Quota and usage figures differ per account.

Corresponds to: V19, FR-1.

D-4 — Media streaming, concurrent (*≈2 min*)

Claim: media streams to standard clients, and concurrently.

Procedure

1. Begin a 1080p stream on the operator laptop.
2. Begin a second stream on a second device.
3. Begin a third stream on a third device (or on the operator laptop in a second browser profile).

4. Observe CPU and network utilisation during the third stream.

```
# Concurrent observation
docker stats --no-stream jellyfin
htop          # or top
```

Required result: three simultaneous streams play without buffering; Jellyfin CPU utilisation remains below full saturation.

Corresponds to: V22, V23, NFR-6.

STREAM SELECTION

Streams are selected from the library such that **Direct Play** is possible. This is stated aloud during the demonstration, together with the reason: Direct Play consumes effectively no server CPU, whereas transcoding is CPU-bound and is the limiting factor on this platform. The 4K transcode boundary is disclosed under §10.

D-5 — Localhost and data custody (*≈1 min*)

Claim: the data plane is on-premises.

Procedure

1. On the operator laptop, open the Nextcloud and Jellyfin sessions.
2. Disconnect the host from any internet uplink.
3. Re-authenticate a new client device and transfer a file.
4. Observe the transfer rate.

Required result: the transfer completes at local network speed with no internet connectivity in use. This is observation D-6 in the verification record.

D-6 — Operation without internet connectivity (*≈2 min*)

SUBSTANTIATES THE CUSTODY CLAIM

This is the demonstration that the system does not depend on a third-party service for its operation.

Procedure

```
# 1. Confirm external reachability before disconnection
```

```

ping -c2 8.8.8.8                # expected: replies

# 2. Physically disconnect the uplink (or disable the interface)
sudo ip link set eth0 down

# (On the assessment network there is no uplink; this step is performed by
# disconnecting the WAN cable at the access point.)

# 3. Re-test all three services

```

Required results

Test	Required result
External reachability	Fails — confirming the disconnection
Nextcloud :8080	Fully functional
Jellyfin :8096	Stream plays normally
CasaOS :80	Renders
New file upload	Succeeds
Background jobs	<code>php occ background:cron</code> completes without error

Corresponds to: V25, C2, NFR-7.

D-7 — Unattended recovery (≈ 2 min)

Claim: the system returns to a defined state without operator action.

Procedure

```

sudo reboot

# Wait for the host to return (~60-90 s from power-on with a wired connection)

# Do not touch the host.

```

Required results

Test	Required result
Reachability after reboot	Automatic, no intervention
All containers	Up , none in Restarting
Nextcloud	Login page loads

Test	Required result
Jellyfin	Dashboard loads
Storage volume	Mounted automatically — confirms <code>nofail</code> and the fstab entry

```
docker ps --format 'table {{.Names}}\t{{.Status}}'
```

Corresponds to: V26, C7.

D-8 — Recovery from backup (*≈4 min*)

✓ SECOND PRIMARY VERIFICATION EVENT

Demonstrates that the recoverability claim is tested rather than asserted.

Procedure

```
# 1. State the recovery model
#   Local snapshots: hourly hardlink snapshots on the data volume
#   Off-site replication: weekly to a removable medium held at a separate location
#   RPO: 1 hour (snapshot interval)   RTO: ≈25 minutes (full rebuild, no data loss)

# 2. Confirm the off-site copy exists and is dated
ls -l /mnt/backup-usb/nextcloud-data/
```

```
# 3. Restore to an independent location
mkdir -p /tmp/restore-verify
sudo rsync -a /mnt/backup-usb/nextcloud-data/ /tmp/restore-verify/

# 4. Compare the restored copy against the live volume
sudo rsync -a --dry-run --delete \
    /tmp/restore-verify/ /mnt/storage/nextcloud-data/
```

Required results

Step	Required result
3	Restoration completes without error
4	No output — the restored copy is byte-identical to the live volume
5	A restored file opens and decrypts with a real user password

Corresponds to: V29, C8, DR-1/DR-2/DR-3.

THE TWO RECOVERY MODELS MUST BE DISTINGUISHED

- **D-7** demonstrates *service* recovery — services return after a restart. The data is untouched.
- **D-8** demonstrates *data* recovery — data is restored from an independent copy.

Conflating the two is a common error in this deployment class. They are different properties, with different failure modes, and this protocol tests both.

D-9 — Declarative reproducibility (*≈2 min*)

Claim: the deployment is declared, not hand-configured, and can be reproduced.

Procedure

```
# 1. Show the complete declaration
cat /opt/picloud/nextcloud/docker-compose.yml
cat /opt/picloud/jellyfin/docker-compose.yml

# 2. Show the recovery path
#   OS card → flash per document 03 §2
#   → install Docker, CasaOS
#   → docker compose up -d    (x2)
#   Data volume is independent of the host.

# 3. Demonstrate teardown and restoration of the service tier
cd /opt/picloud/jellyfin
docker compose down
docker compose ps          # empty
docker compose up -d
docker compose ps          # running
```

Required result: the service tier is fully recreated from the declared files with no manual configuration step.

Corresponds to: C3, C9, P3 in document 01 §4.

D-10 — Health instrumentation (*≈1 min*)

```
picloud-health
```

Required result: the report completes with no `!!!` markers, showing load, thermal and power state, memory, both filesystems, container state, listening ports, and storage health.

Note: this is the same command used for the pre-assessment check (R1–R21). It is included in the sequence to establish that continuous health monitoring is available, not merely a one-off verification.

D-11 — Third-party client compatibility (*≈2 min, conditional*)

Claim: standard mobile and desktop clients interoperate.

Procedure — executed only if the assessor has the application available:

1. Connect the assessor's device to the assessment network.
2. Install or open the Nextcloud application.
3. Sign in with a demonstration account; upload a photograph.
4. Stream a title in Jellyfin.

Required result: both applications connect to `192.168.10.50` and function normally, without any configuration on the host.

If not performed: record as *not performed — precondition unavailable*. The client applications are the same applications used against the commercial services this project replaces, which is the substantive point; it does not depend on which device runs the demonstration.

Sequence summary

ID	Demonstration	Claim established	Verifies	Duration
D-1	Functional overview	System operational; three service classes	V12, V20, V24	2 min
D-2	Encryption, by inspection	Data is ciphertext at rest	V16	3 min
D-3	User isolation	Users are separated	V19	2 min
D-4	Concurrent streaming	Media service under load	V22, V23	2 min
D-5	Localhost data plane	Data remains on-premises	—	1 min

ID	Demonstration	Claim established	Verifies	Duration
D-6	No internet dependency	Independent operation	V25	2 min
D-7	Unattended recovery	Services return without action	V26	2 min
D-8	Restore from backup	Data is recoverable	V29	4 min
D-9	Declarative reproducibility	Build is reproducible	C3, C9	2 min
D-10	Health instrumentation	Continuous monitoring exists	V30	1 min
D-11	Client compatibility (<i>conditional</i>)	Standard clients interoperate	V12	2 min
			Total	≈22 min

6. Evidence artefacts

PURPOSE

The assessment should not depend on the operator's narration. Each claim is supported by a capture that an assessor can inspect afterwards.

Artefact	Captured at	Content
<code>health-report.txt</code>	R1–R21, D-10	Output of <code>picloud-health</code>
<code>encryption-status.txt</code>	D-2 step 1	<code>occ encryption:status</code>
<code>encryption-grep.txt</code>	D-2 step 3	Output of the plaintext search — empty
<code>encryption-hexdump.txt</code>	D-2 step 4	Raw stored bytes of the test file
<code>isolation-log.txt</code>	D-3	Cross-user access attempt and its denial
<code>no-internet.txt</code>	D-6	All service responses with the uplink disconnected
<code>reboot-recovery.txt</code>	D-7	<code>docker ps</code> after unattended restart
<code>restore-diff.txt</code>	D-8 step 4	<code>rsync --dry-run</code> output — empty
<code>compose-declaration.txt</code>	D-9	Both Compose files
<code>verification-sheet.md</code>	§7	Assessor-completed record
Screen recording	Full sequence	Timestamped

CONVENTION

Two of the artefacts are expected to be empty — `encryption-grep.txt` and `restore-diff.txt`. An empty file is the affirmative result in both cases. This is stated explicitly because an empty artefact is otherwise ambiguous.

7. Independent verification sheet

PURPOSE

This sheet is completed by the assessor. **It is not completed by the operator.** It exists so that the assessment record is independent of the project's own reporting.

Part A — Verification of primary claims

#	Claim	Test	Observed result	Assessed
A1	Files are stored on-premises and remain accessible	D-1		☐ Pass ☐ Fail
A2	Stored data is encrypted at rest	D-2 — plaintext string absent from the volume; raw bytes are not readable		☐ Pass ☐ Fail
A3	Encryption is per-user, not a shared master key	<code>occ encryption:status</code> reports per-user		☐ Pass ☐ Fail
A4	Users cannot access one another's data	D-3 — cross-user access denied		☐ Pass ☐ Fail
A5	The system operates with no internet connectivity	D-6 — all services functional with the uplink removed		☐ Pass ☐ Fail
A6	Services recover unattended after restart	D-7 — no operator intervention		☐ Pass ☐ Fail
A7	Data is restorable from an independent copy	D-8 — restored copy identical to the live volume; a file opens		☐ Pass ☐ Fail
A8	The media library cannot be modified by the streaming service	<code>touch</code> inside the container → read-only filesystem		☐ Pass ☐ Fail

#	Claim	Test	Observed result	Assessed
A9	The service tier is reproducible from declared files	D-9 — <code>compose down</code> then <code>up</code> restores without manual steps		☐ Pass ☐ Fail
A10	Host operating system failure causes no data loss	Data resides only on the NVMe; OS is re-provisionable (DR-1/DR-2)		☐ Pass ☐ Fail

Part B — Cost and limit accuracy

#	Statement	Documented value	Value claimed by project	Agrees	Assessed
B1	Capital cost	\$186 / ₹16,500		☐	
B2	Recurring cost	≈\$6 / \$0.50 per month		☐	
B3	Concurrent users	6 mixed workloads		☐	
B4	Concurrent 1080p streams	3–4		☐	
B5	4K transcoding	Not supported		☐	
B6	Storage redundancy	None — single disk		☐	
B7	Uninterruptible power supply	Not installed		☐	
B8	Inbound internet exposure	None		☐	
B9	Multi-factor authentication	Not configured		☐	
B10	Remote access	Disabled		☐	

✓ PURPOSE OF PART B

Part B verifies that **stated limitations match the actual system**. A project that understates its limitations is less reliable than one that states them precisely, because the stated limitations are what make the remaining claims assessable. This section is the mechanism by which that can be checked.

Part C — Assessor notes

Item	Record
Discrepancies observed between documentation and system	
Claims that could not be verified	
Limitations found that are not documented	
Overall assessment	

8. Contingency procedures

PLANNING BASIS

A live assessment depends on venue power, venue networking, and multiple clients. Each is a known failure domain. **The contingency must exist before it is needed**, and each item below is prepared in advance and stored with the deployment.

#	Failure	Pre-prepared contingency	Degraded claim still supported
F1	Host will not power on	Second assembled unit, pre-imaged, same configuration	Full sequence
F2	Host fails during the assessment	Re-image and redeclare from the bootstrap script + two Compose files (≈ 25 min)	All except the data originally on the failed unit
F3	Assessment network unavailable	Laptop-hosted access point with a preconfigured SSID	Full sequence
F4	Visitor/assessor device isolation blocks the demonstration	Secondary SSID with client isolation disabled, on a separate interface	Full sequence, isolation control not demonstrated
F5	Display or output unavailable	SSH-only variant of D-2, D-3, D-6, D-8 — all are executed at the command line	All primary claims; D-1 and D-11 not performed
F6	Insufficient time	Execute D-1 $\rightarrow$ D-2 $\rightarrow$ D-6 $\rightarrow$ D-8 only (≈ 11 min)	O1, O2, O4, O6
F7	Wi-Fi congestion	Wired operator station; host connected by cable	Full sequence

#	Failure	Pre-prepared contingency	Degraded claim still supported
F8	Power interruption at the venue	Host on the USB-C power bank (H8)	Full sequence

MINIMUM VIABLE ASSESSMENT SEQUENCE

D-2, D-6, D-8 — encryption by inspection, operation without internet, and restore from backup. These three cover the two primary technical claims and the principal recoverability claim, and require approximately 9 minutes. **If time or equipment permits nothing else, these three are sufficient to assess the substance of the project.**

CACHED EVIDENCE

A complete screen recording of a prior full run, plus all artefacts from §6, is stored locally. In the event of total demonstration failure (F2 with no replacement unit), this evidence is submitted **and explicitly labelled as a recording of a prior run**. It is not presented as a live demonstration.

9. Security during public assessment

DESIGN INTENT

Public assessment places an unauthenticated audience in contact with a live system. Several controls that are appropriate for a private deployment must be tightened for a public one.

Control	Private deployment	Public assessment	Rationale
Demonstration accounts	Permanent	Regenerated for each assessment session	A credential issued at a public event can be retained; rotating it removes the persistence
Account privileges	Full	Read-only or a scoped sandbox folder	Limits the consequence of any unintended write
Self-registration	Disabled	Disabled, verified as R17	With registration enabled, any connecting device could create an account

Control	Private deployment	Public assessment	Rationale
Media library	Full library	Curated subset	Reduces both exposure and the chance of an unintended title appearing on screen
Data volume	Live	Restored from a verified snapshot immediately before the event	A public event is a realistic trigger for accidental deletion; the snapshot makes recovery immediate
Client isolation	Optional	Enabled on the assessment SSID	Prevents any attendee device from reaching the institutional network or another device
Outbound internet	Available	Disabled	Removes any possibility of data leaving the premises during assessment, and demonstrates O4 by default rather than by arrangement
Shell access	Standard user	Standard user, key-only	An SSH password prompt in a public space is an avoidable exposure

⚡ TWO PRE-EVENT ACTIONS THAT ARE NOT OPTIONAL

1. **Verify self-registration is disabled** (R17). With it enabled, any attendee with the URL can create an account and write to the volume, typically within minutes.
2. **Take a snapshot and confirm the off-site copy is current** (R18, R19). A public event is the most likely occasion for accidental modification. Recovery must be possible before the event, not after.

⚠️ RESIDUAL EXPOSURE, STATED RATHER THAN CONCEALED

Exposure	Assessment
Attendee device attempts authentication against the file service	Mitigated by strong, rotated demonstration credentials and throttling. Not architecturally prevented
Attendee observes the screen and photographs displayed content	Content selection is the control. No confidential data is displayed
Attendee records a URL or a credential	Credentials are rotated per session; URLs are LAN-scoped and become unreachable once the network is withdrawn
Attendee device reaches the host after the event	The assessment network is decommissioned; the host is not reachable from any persistent network

10. Limitations to be disclosed

REQUIREMENT

The following are stated during the assessment, unprompted. **Withholding a known limitation until it is discovered is the single most damaging thing a project of this type can do**, because it invalidates the credibility of every other claim made.

#	Limitation	Disclosure
L1	No storage redundancy	Single disk. A disk failure means total data loss. The only mitigation is the off-site backup in D-8, which protects against loss of the device but not against a failure occurring between replication intervals. Accepted as risk R1 (document 01 §9)
L2	No uninterruptible power supply	A power interruption during a write can lose changes since the last snapshot (RPO $\approx$ 1 hour). Accepted as risk R2. The highest-value single addition to a subsequent revision
L3	4K transcoding is unsupported	Hardware limitation: the VideoCore VII provides hardware decode but not hardware encode. 4K direct play is supported where the client supports the codec. 1080p is the verified envelope
L4	Single node — no failover	No high-availability design. The host is a single device on a single supply at a single location
L5	Multi-factor authentication is not configured	Supported by the platform and is the highest-value addition to the access-control layer. Not implemented in this revision
L6	No remote access	By design (document 01 §6). Any future remote access is to be VPN-only and disabled by default
L7	Server-side encryption is not end-to-end	Protects data at rest against physical and infrastructure compromise. Does not prevent an administrator reading a file during a live authenticated session. End-to-end encryption would disable server-side search, previews, and version history. The trade is explicit, not incidental
L8	Per-user keys mean no password recovery	A forgotten password renders that user's files permanently unreadable. This is a direct consequence of there being no back door. The recovery key exists and its custody is documented (§9.3 of document 03)

#	Limitation	Disclosure
L9	Docker group membership confers root	Accepted for administrative convenience on a single-operator appliance. Not appropriate on a shared host
L10	Setup requires one competent session	≈45 minutes on the critical path, ≈3 hours for the full build. This is a real barrier to adoption and is analysed in document 05 §5. It is the constraint most likely to determine whether a household actually deploys this
L11	Household identifiers are visible on the LAN	MAC address and hardware identifiers are observable by any device on the local network. This is not anonymity and is not claimed
L12	The Raspberry Pi is not the optimal platform for maximum capability	A mini PC is faster and more expandable. It is 6× the capital cost. Recorded in document 02 §13

THE DISCLOSURE PRINCIPLE

Nine of the twelve limitations above are properties of **any** single-node, single-disk, self-hosted deployment, not defects in this implementation specifically. **That is the point being made.** A consumer cloud service solves several of them — redundancy, availability, multi-factor authentication — at the cost of the custody, portability, and recurring expenditure described in document 01. The comparison is not "self-hosting is better"; it is a specific, itemised trade, and the project is explicit about both sides of it.

Summary

Item	Detail
Duration	≈22 minutes full sequence; ≈9 minutes minimum (D-2, D-6, D-8)
Environment	Dedicated access point, no uplink, client isolation enabled, non-overlapping subnet
Preconditions	24 checks, all required to pass before assessment begins
Primary verifications	D-2 encryption by filesystem inspection; D-6 operation without internet; D-8 restore from an independent copy
Evidence	10 artefacts plus a screen recording. Two artefacts are expected to be empty
Verification record	Completed by the assessor, including a section verifying that stated limitations match the actual system

Item	Detail
Contingency	8 documented failure modes, each with a pre-prepared fallback; F2 recovery is $\approx$ 25 minutes with no data loss
Disclosure	12 limitations stated unprompted, of which 9 are inherent to the deployment class rather than to this implementation

Next: [Q5-Everyday-Usability](#) — economic analysis, adoption analysis, migration path, and the assessment of whether this is a viable option for a real household.